

# Problem Statement & Data Cleaning

## 1. Problem Statement

Used and new car pricing varies enormously by brand, engine specifications, and market segment, making it difficult for buyers and sellers to know what a fair price actually is. This project analyzes 11,199 vehicle listings (after cleaning) to identify which factors, engine performance, fuel efficiency, model year, and market positioning, most strongly drive MSRP. The goal is to surface data-backed pricing patterns that can guide more informed buying, selling, and pricing decisions.

## 2. Dataset Description

**Source:** Car Features and MSRP dataset (Kaggle)

Raw size: 11,914 rows × 16 columns

Cleaned size: 11,199 rows × 19 columns (715 duplicate rows removed, 3 columns added)

### Data Dictionary

| Column            | Type     | Description                                                            |
|-------------------|----------|------------------------------------------------------------------------|
| make              | category | Vehicle manufacturer (e.g. Toyota, BMW) — 48 unique brands             |
| model             | category | Specific model name within the make                                    |
| year              | int      | Model year, range 1990–2017                                            |
| engine_fuel_type  | category | Fuel type the engine runs on (e.g. regular unleaded, diesel, electric) |
| engine_hp         | float    | Engine horsepower                                                      |
| engine_cylinders  | int      | Number of engine cylinders (0 for electric vehicles)                   |
| transmission_type | category | Manual, Automatic, Automated Manual, or Direct Drive                   |
| driven_wheels     | category | Drivetrain (front/rear/all/four wheel drive)                           |
| number_of_doors   | int      | Number of doors (2, 3, or 4)                                           |
| market_category   | category | Comma-separated market tags; "Unlabeled" where none existed            |
| vehicle_size      | category | Compact, Midsize, or Large                                             |
| vehicle_style     | category | Body style (Sedan, Coupe, SUV, etc.) — 16 unique styles                |
| highway_mpg       | float    | Highway fuel efficiency, miles per gallon                              |
| city_mpg          | int      | City fuel efficiency, miles per gallon                                 |
| popularity        | int      | Brand-level popularity score (constant per make, not per car)          |

|                       |       |                                                                  |
|-----------------------|-------|------------------------------------------------------------------|
| msrp                  | int   | Manufacturer's Suggested Retail Price, USD — the target variable |
| price_flag_suspicious | bool* | True if msrp == \$2,000 exactly (see Data Quality Flags)         |
| car_age               | int*  | Years since the newest model year in the dataset (2017 – year)   |
| is_electric           | bool* | True if engine_fuel_type contains "electric"                     |

### 3. Data Cleaning Summary

#### 3.1 Duplicates

715 fully duplicate rows were identified and removed (11,914 → 11,199 rows). These were exact row-for-row repeats that would have double-counted specific listings in later statistical analysis.

#### 3.2 Missing Values

Each column was handled individually rather than with a blanket drop or fill, since a one-size-fits-all approach would either discard usable data or introduce values that don't reflect reality.

| Column            | Missing                | Strategy                                                                      | Reasoning                                                                                     |
|-------------------|------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| engine_fuel_type  | 3                      | Filled with mode                                                              | Too few rows to justify a more complex approach                                               |
| engine_hp         | 69                     | Median HP of same make+model (overall median as fallback)                     | HP varies enormously by model; a flat average misrepresents both economy and performance cars |
| engine_cylinders  | 30                     | Electric vehicles set to 0 (factual); rest filled with same make+model median | Electric vehicles genuinely have no cylinders, a known fact, not a guess                      |
| number_of_doors   | 6                      | Filled with mode of same vehicle_style                                        | Door count is driven by body style more than by brand                                         |
| market_category   | 3,742 (31%)            | Filled with "Unlabeled" as its own category                                   | Too large a share to drop; missingness is itself meaningful                                   |
| transmission_type | 12 (as text "UNKNOWN") | Replaced with mode                                                            | Hidden missing-value token found via a systematic scan across all categorical columns         |

#### 3.3 Data Type Corrections

- engine\_cylinders and number\_of\_doors converted from float to int, now that missing values are resolved.
- All text/categorical columns converted to pandas category dtype for memory efficiency and faster grouping operations in later analysis.

#### 3.4 Outliers & Data Quality Flags

Two genuine data quality issues were identified through systematic inspection, comparing each column's 75th percentile to its maximum value, then investigating any column where that jump was disproportionately large, rather than by assumption.

### **MSRP placeholder value.**

A frequency check (`value_counts()`) revealed that exactly \$2,000 appears 748 times in the cleaned dataset, over 30 times more often than the next most common price, disproportionately on vehicles from the early 1990s. This is almost certainly a placeholder value rather than a genuine price. Rather than deleting these rows (which would silently shrink the dataset) or trusting them at face value (which would distort any price-based analysis), they were flagged, not removed, via a new `price_flag_suspicious` column. Analysts in later stages should exclude or caveat these rows when computing price-based statistics.

The high end of the MSRP range (up to ~\$2.07M) was also checked and found to be legitimate, real exotic and luxury vehicles, not data errors. No action was taken on the high end, though a log scale is recommended for any visualization of price distribution given the scale of this legitimate variance.

### **highway\_mpg data entry error.**

The same systematic check flagged `highway_mpg`, whose maximum value (354) was nearly 12 times its 75th percentile. Investigation showed this was almost entirely explained by legitimate electric vehicles, which report unusually high MPG-equivalent ratings except for one row, a 2017 Audi A6, where `highway_mpg` = 354 was paired with a normal `city_mpg` of 24. Real electric vehicles show both figures elevated together; this mismatch confirmed it was a data entry error rather than a legitimate reading. The value was corrected using the median `highway_mpg` of the same make and model.

## **3.5 Feature Engineering**

Two features were added to support later analysis stages:

- `car_age` : years since the newest model year in the dataset, often more useful for trend analysis than a raw calendar year.
- `is_electric` : boolean flag derived from `engine_fuel_type`, saving later stages from recalculating it.